

Black Holes – The Immortal Live-Givers

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Abstract

This preprint develops the Universal Quantum Foam Hypothesis as an ontological framework for a field mechanical interpretation of gravity, dark matter, dark energy, and the dynamics of compact objects. Physical space is described as a continuous elastic field medium whose local stress states determine observable gravitational phenomena. Matter and radiation emerge as bound and free organizational modes of the same medium. Dark matter effects are interpreted as persistent stress gradients of the field, while dark energy corresponds to large scale relaxation behavior. Within this framework, the formation and merger of black holes are modeled as saturated regime transitions of the medium. Instead

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of point like singularities, structured and elastically limited field nodes arise, whose coupling generates jets, matter outflows, and large scale structure formation. The approach does not replace established equations but provides a unified field mechanical interpretation of observations across scales.

1 Black Holes as Regime Transitions of the Quantum Foam

Within the framework of the Universal Quantum Foam Hypothesis, a black hole is not primarily understood as an “object” that exists within an otherwise empty space, but rather as a local boundary state of an active medium. Physical space is, in this view, a continuous quantum foam, that is, a field mechanical carrier medium that not only allows excitations, bindings, and relaxations, but itself appears as these states. Matter is a bound organizational form of this medium, radiation a free or freeing organizational form, and gravitation is the geometric response of the medium to bound, stable, or metastable field states. Crucially, gravitation is not regarded as an additional “force” alongside other forces, but as an expression of the tension and curvature organization of the medium itself. The medium carries tension, it distributes tension, and it responds to tension with geometry.

Against this background, the formation of a black hole is not a transition into a new substance, but a transition into a different operating regime of the same field. This is precisely what is meant by a regime change. In the UQSH, a regime change does not mean that physics “stops” or that infinite quantities become real at a certain point, but rather that a particular type of field response loses its validity and is replaced by another. As long as the medium can still elastically carry local stress in the form of increasing curvature and local densification, it responds with gradual

geometry. However, once a region is reached in which additional loading can no longer be realized as further local deepening, the character of the field response changes. Additional stress is then no longer translated into point like sharpening, but into redistribution, lateral expansion, shielding, collective reorganization, and preferred relaxation channels. The regime change is therefore not a mysterious point, but the physical consequence of the medium’s finite load bearing capacity.

This also clarifies why the elasticity limit should not simply be equated with the size of a black hole. The elasticity limit is first and foremost a property of the medium itself, namely the limit up to which a local organization of tension or curvature can be supported without tipping into a different response mode. The size of a black hole, by contrast, describes the spatial extent over which this boundary state is organized and how far the associated saturation zone extends outward. In this reading, a larger black hole is not “more at the limit” because it is larger, but rather it is spatially more extended in a state that operates closer to the boundary response. It carries the boundary organization across larger scales, whereby the environment is structured differently, the gradients become flatter or more extended, and the manner of coupling to accretion, magnetic fields, and radiation shifts. The confusion easily arises because in everyday intuition “larger” appears “more extreme.” In the UQSH, however, “extreme” is primarily a regime concept. It refers to the type of field response, not merely to a number or a radius.

One may illustrate the transition intuitively, without trivializing it, by distinguishing between three levels. First, there is the local field tension, that is, how strongly the medium is geometrically constrained in a given region. Second, there is the ability of the medium to still accommodate additional tension as further local deepening. Third, there are the alternative response forms once this ability is exhausted. A black hole marks the situation in which the second level fails and the third dominates. This

does not mean that no dynamics take place anymore, but that dynamics become increasingly internally organized, channeled, and transformed into controlled forms of relaxation. In this sense, the event horizon in the UQSH is less a place where “something disappears,” but rather a threshold at which the transmission of free field motion into open propagation is replaced by a predominantly internally bound organization. The invariance of the limiting dynamics remains preserved, but it no longer appears as a freely propagating mode, rather as a bound, saturated field organization.

This perspective is important for understanding the later fusion process of maximally saturated systems in a consistent way. When two supermassive black holes approach each other, it is not simply two point like masses that meet, but two extended tension landscapes whose outer zones already operate in a boundary regime. The “early coupling” is then not an additional assumption, but follows directly from the fact that the medium has no sharp object boundaries and that boundary zones react sensitively to additional loading. The intermediate region becomes dynamically relevant because it becomes the place where the two tension profiles overlap without the medium being able to become arbitrarily “deeper” there. It is precisely at this point that the mechanism begins which is later described as the pressure node, elastic repulsion moment, barrier contact, and shared saturation membrane. In the UQSH, these terms are not metaphors for objects, but designations for response forms of the medium in a nonlinear boundary regime.

2 Conceptual Foundations and Field Mechanical Quantities

In the following, physical space is not understood as an empty background or as a purely geometric stage, but as a continuous, dynamic field medium. This medium, referred to within the framework of the UQSH

as the quantum foam, forms the carrier of all physical processes. Matter, radiation, and gravitation do not act within this space, but are themselves states of this medium.

The quantum foam is locally deformable, elastic, and responsive. Every input of energy, every radiative excitation, and every bound material structure modifies its state of tension. Physical processes therefore do not correspond to the motion of objects through empty space, but to reorganizations of the field. Motion is field reconfiguration, radiation is propagating field excitation, and matter is a stably bound field state.

Within this framework, gravitation does not appear as an independent force with a separate coupling constant, but as a geometric consequence of these tension states. Stable or long lived structures imprint the medium, generate gradients and curvatures, and other structures follow these predefined paths. Gravitational action is therefore not an additional interaction, but the dynamic response of the field to its own organization.

In this view, a black hole does not represent a singular point of infinite density, but a boundary state of the medium. When the local field tension becomes so large that further compression is no longer possible, the quantum foam reaches its maximum elastic load capacity. Beyond this point, the medium no longer responds with additional deepening, but with redistribution, lateral expansion, and collective reorganization. This transition marks a regime change in the field dynamics, as discussed previously.

This conceptual classification forms the foundation for the following presentation. The formation, coupling, and merger of black holes is not treated as a collapse like formation of singularities, but as the structured dynamics of an elastic medium whose tension states reorganize, couple, and give rise to new large scale order.

3 Merger of Supermassive Black Holes

3.1 Early Coupling of Maximally Saturated Systems

Within the framework of the UQSH, supermassive black holes represent maximally saturated field states. They mark local boundary regions in which the elasticity of the quantum foam is nearly completely exhausted and additional local compression can no longer be compensated by further curvature, but only by redistribution or expansion of field tension. Each of these nodes is therefore surrounded by an extended saturation zone in which the medium already operates close to its maximum sustainable tension.

When two such systems approach one another, their interaction does not begin only at the moment of geometric or horizon contact. Even at large distances, their tension profiles begin to overlap. Because the field is continuous and possesses no sharp object boundaries, not two isolated cores meet, but two large scale tension landscapes. The outer boundary regions of high field tension react particularly sensitively to additional loading. As a result, an early field mechanical coupling arises, in which the respective saturation barriers mutually modulate each other. The medium reacts to the superposition of two maximal states as a local overload situation and begins to adapt structurally in advance.

In the region between them, the tension gradients of the two systems cannot simply superpose linearly or reinforce each other additively. In this regime, the quantum foam does not behave like an ideal, infinitely compressible continuum, but like a medium with finite elasticity and with a locally maximal sustainable field tension. As soon as two strongly saturated tension zones approach one another, the field between them rapidly reaches a state in which additional loading can no longer be absorbed through further local curvature. A direct deepening of the field

structure would exceed the elastic limits of the medium and is therefore field mechanically inadmissible.

Instead, the medium responds through reorganization. The injected energy is not converted into stronger point like compression, but into collective, spatially distributed deformations. Field lines are deflected laterally, curvatures spread outward, and the tension energy distributes itself over a larger volume. This process does not correspond to a local sharpening, but to densification through concentration and mutual interlocking of the gradients. The effective stiffness of the field increases, the geometric structure becomes more compact, and the dynamics slow down because the medium increasingly builds resistance against further compression.

In this way, a common region of elevated tension density emerges between both systems, in which the field responses of both sides overlap and mutually reinforce each other. This region is neither clearly attributable to one of the two cores nor to be understood as a separate object. Rather, it is an emergent field structure that arises solely from the interaction of the two tension landscapes. Its existence is not based on additional matter, but on the geometric organization of the medium itself. The intermediate region thus becomes an energetically dominant node in which a considerable portion of the total field tension of the configuration is stored.

This so called pressure node functions as the central coupling element of the system. It determines how quickly the cores can approach one another, how strongly they are decelerated, and along which paths energy and momentum are redistributed. The further dynamics are therefore not primarily governed by the individual objects, but by the common field structure that has formed between them. The interaction of two maximally saturated black holes is thus from the outset no longer a two body dynamics, but the evolution of a continuous tension field in which the intermediate region itself becomes the decisive physical quantity.

As soon as the tension stored in the intermediate region exceeds the local loading of the individual cores, the character of the field response changes fundamentally. The medium no longer reacts passively to the approach, but enters an active, restoring state. Whereas previously additional energy could still be converted into collective reorganization and densification, this margin is now also exhausted. The intermediate region reaches a higher effective tension density than the adjacent cores themselves. As a result, the local gradient of field curvature reverses. The steepest energetic region no longer lies in the centers of the two objects, but in the space between them.

In an elastic medium, such a gradient inevitably implies a counter motion. Field tensions always tend to reduce themselves and to escape into less loaded regions. The highly compressed intermediate region therefore no longer acts attractively, but repulsively. The field reorganizes in such a way that stored tension can flow laterally or outward. This redistribution generates an effective force component that increases the distance between the two cores or at least significantly slows their further approach. Physically, this appears as recoil or as a temporary decoupling of the systems.

This repulsion moment is not an additional or exotic mechanism, but the direct consequence of the finite elasticity of the quantum foam. Instead of accumulating physically unrealistic tensions, the field releases part of the stored energy and converts it into macroscopic motion or radiation. The dynamics thus resemble less a gravitational collapse and more a mechanical spring process in which approach and restoration alternate.

Consequently, the interaction does not proceed monotonically. The systems do not move steadily toward one another, but enter a state of repeated approach and rejection. Phases of increased compression alter-

nate with phases of partial relief. The trajectories become oscillatory, intermittent, and increasingly chaotic. Part of the stored energy is dissipated into radiation, magnetic fields, or directed outflows, whereby the overall system gradually relaxes. The approach therefore does not occur as an immediate fall into a singularity, but as a temporally extended, field mechanically moderated process in which the medium itself assumes the role of a regulating and stabilizing agent.

3.2 Mechanics of the Merger

With progressive approach, the two maximally saturated field nodes do not immediately enter into direct core contact. Initially, only their outer saturation zones interact. The previously formed tension barriers gradually overlap and form a common boundary region in which the field geometries of both systems couple with one another. This barrier contact does not represent an abrupt collision, but a continuous adjustment of two highly saturated tension profiles. The medium does not react discretely, but elastically. Tensions are distributed, smoothed, and laterally redirected. In this way, a connected, surface like transition zone emerges, which can be understood as a shared saturation envelope.

This envelope functions as a collective protective structure of the quantum foam. It absorbs local overloads, distributes pressure over larger regions, and prevents individual points from becoming physically unrealistically compressed. The approach therefore does not proceed in a collapse like manner, but oscillatory and dissipative. Energy is continuously dissipated into radiation, turbulence, and large scale field reorganization. The process is structure forming rather than destructive. Instead of an abrupt breakdown, an increasingly coherent coupling of both systems arises. The dynamics resemble less an explosion and more a mechanical locking of two elastic structures that gradually form a common geometry.

In the further course, the intermediate region between the two nodes transitions into a structured transitional state. Since additional local deepening is no longer possible, this region loses its function as a separating zone. The stored tension is redistributed and discharged into directed relaxation channels. The intermediate region does not collapse in the sense of a singularity, but reorganizes itself as a new, connected field structure. Energy is preferentially transported along stable axes, often in the form of jets or large scale radiation flows. The merger thereby appears ordered and mechanically guided.

At the same time, the internal geometries of both cores adapt to the shared boundary surface. Since further radial compression is excluded, the response occurs laterally. The previously nearly spherical tension profiles are stretched sideways and flattened. The cores broaden within the contact zone and reduce local curvature peaks. This lateral deformation minimizes gradients and facilitates the formation of a stable common configuration. The behavior resembles that of two strongly tensioned elastic bodies which, under pressure, can no longer be compressed further and therefore yield sideways.

In parallel, an energetic synchronization sets in. The previously independent rotations, oscillation modes, and internal field rhythms begin to influence one another. Phase differences are reduced through dissipation, characteristic frequencies converge, and a collective resonance emerges. This adjustment reduces shear forces and prevents chaotic instabilities within the contact region. The medium strives toward a state of minimal relative motion in which both systems can respond as a coherent unit. The merger is thus not forced, but prepared.

Only after this geometric and dynamical adjustment has taken place does the actual core contact occur. This contact likewise does not take place point like, but surface like. Two highly saturated field states do

not touch like rigid objects, but merge across an extended zone of shared tension. Along the contact line, a ring shaped tension node forms, which functions as the core seam of the new system. This seam stabilizes the emerging overall structure, channels remaining tensions away, and defines the axis of the unified field node. The transition from a double to a single system therefore appears not as a singular collapse, but as a continuous reorganization of an elastic medium that autonomously adapts its geometry and transforms itself into a new, stable order.

3.3 Structure Formation and Scale Coupling

Well before the actual core contact, pronounced radiation mechanical activity sets in. The increasing coupling of the tension zones leads to stored field energy no longer remaining localized but being discharged along preferred relaxation directions. The quantum foam has no possibility for further compression, so excess load is converted into directed propagation modes. These manifest as early, still unstable radiation streams emerging from the shared boundary region. Such pre jets are not secondary byproducts of the fusion, but direct consequences of the elastic field response during the approach. The medium already begins, at the stage of coupling, to dissipate energy and actively reorganize its surroundings.

With progressive synchronization of the two systems, these directed outflows intensify. Tensions are no longer dissipated diffusely, but concentrated along stable axes. A pull push dynamics forms, in which field energy is drawn into the contact zone on the one hand and expelled along collective channels on the other. The merger thus acts like a pump that continuously transports energy, momentum, and radiation from the intermediate region outward. This phase is characterized by high dissipation, strong magnetization, and pronounced anisotropy. Jets therefore do not arise only as a consequence of a completed merger event, but accompany and structure the entire process.

At the stage of maximal coupling, these outflows reach their greatest stability and extent. The radiation becomes strongly collimated, remains coherent over large distances, and carries significant fractions of energy out of the system. Such large scale jets do not merely act as channels of energy loss, but actively intervene in the environment. They compress matter, generate shock fronts, amplify magnetic fields, and create preferred condensation zones. Filaments, dense nodes, and early structure formations appear in this view not as random byproducts of gravitational instability, but as direct field mechanical consequences of directed relaxation flows. Jets thus assume an organizing function within the cosmic medium.

After synchronization and contact formation are completed, the coupled system transitions into a shared, maximally saturated state. From two separate tension centers, a single coherent field node emerges with larger spatial extent and higher collective stability. This supernode does not represent a singular point, but a large scale, saturated tension region of the quantum foam. Its effect is not based on infinite compression, but on a persistently organized field geometry that carries and distributes tensions across wide areas.

The material expelled during coupling does not remain waste. Matter and radiation released from the system along the jets form new condensation seeds. Densifications arise, cool, and stabilize into clouds, clumps, rings, or filamentary structures. Under suitable conditions, compact tension centers may again develop from these, themselves transitioning into strongly bound states. In this way, smaller black holes or dense gravitational nodes emerge, which in turn serve as anchor points for further structure formation. The products of the merger thus immediately become starting points of new organization.

From this perspective, the dynamics of maximally saturated systems do not appear as an isolated end stage, but as part of a continuous cycle of field mechanical restructuring. Compression leads to tension buildup, tension buildup to directed discharge, discharge to new matter distribution, and this in turn to renewed compression. In this context, black holes do not primarily act as destructive sinks, but as transformers and organizers of the medium. They collect, structure, and distribute energy and matter across large scales.

The resulting supernode remains as a stable component of its surroundings. It smooths local tensions, bundles field lines, and stabilizes neighboring filaments. In this sense, such objects function as persistent anchors of large scale architecture. Gravitation here appears not merely as the contraction of mass, but as the expression of a medium that actively orders its own geometry. Black holes therefore mark not endpoints of physical development, but central nodes of continuous structure formation.

With the formation of stable jets and directed energy channels, the fusion process is not limited to pure field reorganization. In the strongly compressed and highly energetic boundary regions between the coupled tension zones, conditions arise under which radiation is not only transported, but converted into bound states. Extreme field compression, high photon energy density, and strong magnetic structuring favor pair production processes. From intense radiation, electron positron pairs locally emerge, manifesting as short lived but real material excitations of the medium. In the UQSH, these particles are not foreign entities, but condensed field states, that is, temporary binding forms of the same quantum foam that also carries the gravitational structure.

Some of these pairs annihilate rapidly and feed energy back into the radiation dynamics. Others couple to existing plasmas, magnetic fields, and shock fronts and are thereby stabilized. In this way, real matter

components arise directly from the relaxation dynamics of the field. The fusion of maximally saturated nodes thus acts not only as an energy drain, but simultaneously as a source of matter. Pure field tension is converted again into mass through radiation and pair formation. This newly formed matter subsequently influences the tension state of the medium itself, strengthens local gradients, and actively intervenes in the further dynamics of the environment. Structure therefore emerges not only through rearrangement of existing matter, but through continuous conversion between field tension, radiation, and bound states.

Accordingly, the merger does not appear as a one sided accumulation process in which ever more mass is collected in a single object. Rather, it acts as a transformation node that absorbs tension, radiates energy, and simultaneously generates new material structures. A growing supernode therefore does not store arbitrarily more substance, but attains increasingly stronger field mechanical stiffness with increasing size. The saturation limit of the medium shifts progressively outward with the expansion of the node. Additional load no longer leads to further compression of the center, but preferentially to lateral expansion, enhanced jet formation, and increased matter outflow.

For this reason, unlimited hierarchical merging into a single all encompassing object within a filament is not to be expected. The larger a node becomes, the less efficient further growth becomes. A growing tension region reacts increasingly elastically and diverts additional energy and matter outward ever earlier. The system asymptotically approaches a maximally sustainable state in which further mergers primarily lead not to mass increase, but to enhanced reorganization of the surroundings. Large nodes stabilize filaments, shape structures, and feed matter back into their environment rather than accumulating everything.

In the formal description, this boundary regime is characterized by the

parameter Qu_{Horizon} . It does not denote an absolute maximum value in the sense of a universal constant, but the greatest tension organization of the quantum foam realizable within a given regime. When this state is reached, the medium responds almost entirely restoratively. Further load primarily produces expansion and radiation instead of additional compression. The supernode then functions as a saturated anchor state rather than an indefinitely growing sink.

Black holes therefore appear in this view not as final repositories of matter, but as regulatory elements of an ongoing cycle. Field tension is collected, converted into radiation, partially transformed into new matter, and subsequently redistributed on large scales. Maximum saturation prevents both global collapse and infinite accumulation. Structure emerges through balance between compression and relaxation. The quantum foam organizes itself.

3.4 Final Parsec

In established astrophysics it has long been known that two supermassive black holes, following a galaxy merger, initially approach each other relatively quickly but then can seemingly stall at separations of about one parsec. At this stage, stars have largely been scattered out of the intermediate region, gas is often too thin to generate sufficient friction, and gravitational waves are still too weak at these distances to efficiently transport energy away. In the standard picture, a clear mechanism for further loss of energy and angular momentum is therefore lacking. This so called final parsec problem thus describes a phase in which two massive objects are gravitationally bound yet nevertheless approach each other only very slowly.

Within the field-mechanical framework of UQSH, this situation presents itself differently. Two maximally strained field nodes do not simply

exist as isolated masses in empty space. As soon as the two strain fields meet, this area begins to reorganize itself. The field tension stored there is continuously redistributed and dissipated. This process can be visualized as if the medium were gradually clearing the space between them. Energy is not stacked up, but rather translated into motion. It escapes in the form of radiation, local oscillations, and directed outflows.

The quantum foam thus acts like an intrinsically damping medium. It absorbs excess tension and releases it again. In this way, both systems continuously lose energy and angular momentum even before gravitational waves dominate. What appears as a standstill in the standard model is, in the UQSH picture, an active transitional phase. The medium itself ensures that the approach is not blocked but continues slowly, oscillatory, and dissipatively. The so called final parsec is therefore not a barrier, but a phase of enhanced field mechanical coupling in which the space between both systems becomes increasingly structured and energetically relieved.

4 Black Hole as a Radial Stellar Scale

Spectral field structure of accretion disks and radiation nodes within the framework of the UQSH

Within the framework of the UQSH, radiation is not an emission of discrete particles separate from matter, but the immediate relaxation of locally stored field tension in the quantum foam. Every observed electromagnetic frequency represents the characteristic response rate with which a previously bound or compressed field state converts its deformation back into free motion. The wavelength is therefore not an external label, but a direct measure of the local tension state of the medium.

This perspective enables a field mechanical reinterpretation of radiation in

the environment of black holes. A black hole itself emits no radiation from its interior. The observed signatures arise in the surrounding accretion disk, in magnetized coronae, as well as in jets and shock regions. In all these areas, matter and radiation are coupled to the quantum foam in extreme tension states. The closer such a region approaches the saturated core, the more density, curvature, and field mechanical pre tension of the medium increase. The stored energy cannot remain locally bound without limit and is therefore released again in the form of high frequency relaxation processes.

This produces a continuous radial tension gradient. In outer zones with lower density and weak field deformation, long wavelength emissions in the radio and infrared range dominate. In intermediate regions, the characteristic relaxation frequency shifts into the optical and ultraviolet range. In the innermost zones, where the field is maximally compressed and strongly coupled, X ray and gamma emissions occur. The spectrum therefore does not reflect different physical mechanisms, but different intensities of the same field mechanical process.

In this sense, the accretion disk can be understood as a radial sequence of radiation nodes. Each radius corresponds to a specific local saturation state of the quantum foam and thus to a characteristic frequency window. Phenomenologically, this stratification resembles a sequence of increasingly hotter stellar states. Outer regions behave spectrally similar to cool stars with dominant infrared radiation, further inward solar like emissions appear, while the innermost zones exhibit properties of extremely compressed, high energy stellar plasmas. An active galactic nucleus thus appears not as a point source, but as a continuously structured field body whose radiation directly maps the local tension distribution of the medium.

This interpretation is consistent with the basic principle of the UQSH

according to which radiation always arises isotropically as a spherical field response. This mechanism remains valid even in the vicinity of black holes. The observed directional dependence of jets or radiation beams is not a violation of isotropy, but a consequence of geometric and magnetic channeling. The field itself responds locally in a spherical manner, yet global gradients, rotations, and guiding structures direct the released energy preferentially along certain axes. The anisotropy therefore arises from the organization of the medium, not from an intrinsically directed emission.

Overall, the spectral diversity of active galactic nuclei appears within the UQSH not as the sum of separate processes, but as a direct mapping of a continuous tension profile of the quantum foam. The electromagnetic spectrum thus functions as a diagnostic instrument of local field saturation. Every frequency is the signature of a specific state of the medium. Accretion disks thereby become visible field gradients, and black holes become central organizational nodes where extreme tension states accumulate, reorganize, and relax into radiation.

The spectral distribution of radiation emitted by accretion disks in this framework allows a direct field mechanical diagnostic of the central node state. Since radiation in the UQSH is not understood as an independent energy carrier but as a relaxation form of locally stored field tension, each observed frequency corresponds to a characteristic rate at which the quantum foam releases stored deformation. The electromagnetic spectrum therefore does not represent a thermal property in the classical sense, but a map of tension gradients within the medium.

With decreasing distance to the central black hole, the local field curvature increases continuously. Matter in the accretion disk is forced into increasingly dense and faster orbital states, thereby deforming the quantum foam more strongly and energetically saturating it. The resulting relaxa-

tion processes occur at higher rates, which manifest as shorter effective wavelengths. Long wavelength emissions therefore originate preferentially in outer, weakly stressed regions, while short wavelength emissions up to X ray and gamma emissions arise from regions of maximal field compression near the inner disk edge. The spectrum is thus directly proportional to the local field tension.

Crucially, this maximum frequency is not universal, but depends on the size and structure of the central field node. A compact, low mass node generates steep tension gradients, high local saturation, and correspondingly hard emission. An extended, massive node distributes the same field tension over larger volumes, so that local gradients remain flatter and the spectrum appears overall softer. Large systems therefore radiate predominantly in the optical and ultraviolet range, whereas smaller systems exhibit relatively stronger high energy components. The apparent temperature of the disk is in this interpretation merely an indirect expression of the underlying field tension.

From this relation it follows that the observed spectrum represents a direct measure of the physical size of the black hole. The highest stable emission frequency marks the innermost attainable relaxation state of the medium and thus the maximum local saturation level. In combination with line widths, Doppler profiles, and temporal variability, the radial structure of the accretion disk can thereby be reconstructed. Spectral diagnostics thus becomes a mapping of the field mechanical geometry itself. Not the mass alone is measured, but the way in which the quantum foam is organized and tensioned by the central node.

5 The Immortal Life Givers

In this picture, black holes do not appear as dark objects, but as particularly strong radiation nodes whose surroundings continuously release a spectrum of differently bound relaxation states. The accretion disk forms a radial sequence of increasing field tension, comparable to a continuous series of differently dense stellar states within a single system. Each observed wavelength corresponds to a specific level of this binding. The electromagnetic spectrum thereby becomes the immediate signature of the internal structure of the field node and enables a consistent, scale independent determination of its properties without recourse to additional, unobservable components.

A central characteristic of the dynamics described within the framework of the UQSH is its pronounced scale continuity. The same field mechanical principles that lead to the formation of short lived excitations, pair production, and local relaxation at microscopic lengths continue on astrophysical and cosmic scales as gravitation, jet formation, and structure forming processes. There exists no categorical separation between micro and macro, but only different organizational forms of the same medium. Local condensations generate tension gradients, tension gradients lead to directed energy release, and energy release stabilizes or produces new bound material states. What appears in the laboratory as pair production or plasma emission manifests on the galactic scale as jets, filaments, or star forming regions. Within this continuity, black holes represent no exception, but rather the most extreme expression of the same field mechanical principle.

During the merger of maximally saturated nodes, field tension first concentrates in the intermediate region, reorganizes elastically, and is ultimately discharged in the form of strong radiation and collective outflows. In these highly energetic regions, electron positron pair formation and further

bound states arise that can be stabilized as real matter. Field tension is thereby directly converted into new structural building blocks. Jets, filaments, and matter islands are in this view not secondary byproducts, but the primary channels through which the medium transfers stored energy back into organized form. Black holes therefore act not only as gravitational sinks, but simultaneously as sources of new structure.

With increasing size, however, the local stiffness of the field grows. The saturation limit shifts outward, additional load is increasingly diverted laterally at an early stage, and further growth becomes inefficient. Large nodes therefore tend more strongly toward energy release and reorganization of their surroundings than toward unlimited accumulation of mass. This prevents a hierarchical concentration of all matter into a single object. Instead, a dynamic equilibrium emerges between compression, discharge, and renewed structuring. The parameter $Qu_{Horizon}$ describes this maximally sustainable tension state as a regime boundary at which additional compression primarily transitions into expansion and radiation.

At the same time, black holes are not absolutely stable final states. Over very long timescales they lose energy through Hawking radiation. This evaporation, however, proceeds extremely slowly and is typically surpassed on cosmic scales by repeated interactions, accretion processes, and mergers. A black hole therefore does not remain isolated, but continuously undergoes phases of coupling, fusion, energy release, and structural reorganization. It does not simply disappear, but transforms continuously within the field.

From this field mechanical perspective, black holes appear less as endpoints of development and more as long lived centers of organization within a cyclical process. They accumulate tension, structure matter, release energy, and generate new bound states from which stars, gas clouds, and smaller compact objects subsequently emerge. Across many

generations of mergers and relaxations, they become persistent nodes of cosmic architecture. Their identity is not tied to a specific ensemble of particles, but to the enduring field structure itself. In this sense, they can be understood as functionally immortal. Not as immutable objects, but as recurring, transforming patterns of the same medium that generate and maintain structure over arbitrary timescales.

Black holes thus appear as the longest lived and most effective organizers of the universe. They do not terminate development, but drive it forward. They are not cosmic endpoints, but regenerative nodes in the continuous cycle of field tension, radiation, and matter.

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